

# Angles, polygons, congruence and vectors - Set 2 - Solutions

Specification M5.1-M5.6, M5.10, M5.14, M5.18, M5.19.

## Answer key

| Question | Answer | Topic                                    |
|----------|--------|------------------------------------------|
| 1        | D      | Angles, polygons, congruence and vectors |
| 2        | D      | Angles, polygons, congruence and vectors |
| 3        | E      | Angles, polygons, congruence and vectors |
| 4        | C      | Angles, polygons, congruence and vectors |

## Worked solutions

### Question 1 GEO-04 · Angles, polygons, congruence and vectors · spec M5.19

In triangle  $OAB$ ,  $\vec{OA} = \mathbf{a}$  and  $\vec{OB} = \mathbf{b}$ .  $M$  is the midpoint of  $AB$  and  $P$  is the point on  $OM$  with  $OP : PM = 2 : 1$ . Express  $\vec{AP}$  in terms of  $\mathbf{a}$  and  $\mathbf{b}$ .

**Answer: D**

$$\vec{OM} = \frac{1}{2}(\mathbf{a} + \mathbf{b}), \vec{OP} = \frac{2}{3}\vec{OM} = \frac{1}{3}(\mathbf{a} + \mathbf{b}). \vec{AP} = \vec{OP} - \mathbf{a} = \frac{1}{3}\mathbf{b} - \frac{2}{3}\mathbf{a}.$$

### Question 2 GEO-06 · Angles, polygons, congruence and vectors · spec M5.14

A ship sails from  $P$  on a bearing of  $060^\circ$  for 10 km to  $Q$ , then on a bearing of  $150^\circ$  for 10 km to  $R$ . What is the bearing of  $P$  from  $R$ ?

**Answer: D**

The turn at  $Q$  is  $90^\circ$  and  $PQ = QR$ , so triangle  $PQR$  is right-angled isosceles and  $PR$  makes  $45^\circ$  with  $QR$ . Bearing of  $R$  from  $P$  is  $060 + 45 = 105^\circ$ ; the back bearing is  $105 + 180 = 285^\circ$ .

### Question 3 GEO-11 · Angles, polygons, congruence and vectors · spec M5.19

$\vec{AB} = 3\mathbf{p} + 2\mathbf{q}$  and  $\vec{BC} = k\mathbf{p} - 4\mathbf{q}$ , where  $\mathbf{p}$  and  $\mathbf{q}$  are not parallel. Given that  $A$ ,  $B$  and  $C$  are collinear, find  $k$ .

**Answer: E**

$$\text{Collinear means } \vec{BC} = \lambda\vec{AB}: -4 = 2\lambda \Rightarrow \lambda = -2, \text{ so } k = 3\lambda = -6.$$

### Question 4 GEO-15 · Angles, polygons, congruence and vectors · spec M5.2

A regular hexagon and a regular octagon share a common side. What is the size of the angle between their two other sides that meet at one end of the common side (the angle outside both polygons)?

**Answer: C**

$$\text{Angles at the shared vertex: } 120^\circ (\text{hexagon}) + 135^\circ (\text{octagon}) + x = 360^\circ, \text{ so } x = 105^\circ.$$